pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Volatility Clustering and GARCH

WEEK 2 · DAY 2

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Today: six steps

Six steps

1. See that the size of a change has memory
2. Read a GARCH(1,1): one recursive formula
3. Draw rolling, EWMA, and GARCH volatility
4. Read every setting of arch_model
5. Score walk-forward against simple rules
6. Say what changed from day 5

The question

The 10 Yr's change had no memory in its level and much in its size. How much does that buy in a walk-forward score? Sections 7.1 to 7.4 describe 2 Jan 2015 to 30 Dec 2022 (2,001 dates); 7.5 scores.

The finding is a contrast with day 5, and a modest one.

Names for a model's output

A GARCH's output is a fitted volatility in bp, or the model's value for that date. A miss is a residual. The baseline for a size:

Allowed names

fitted volatility in bp

the model's value for that date

residual

Never called

a forecast of volatility

an expected move

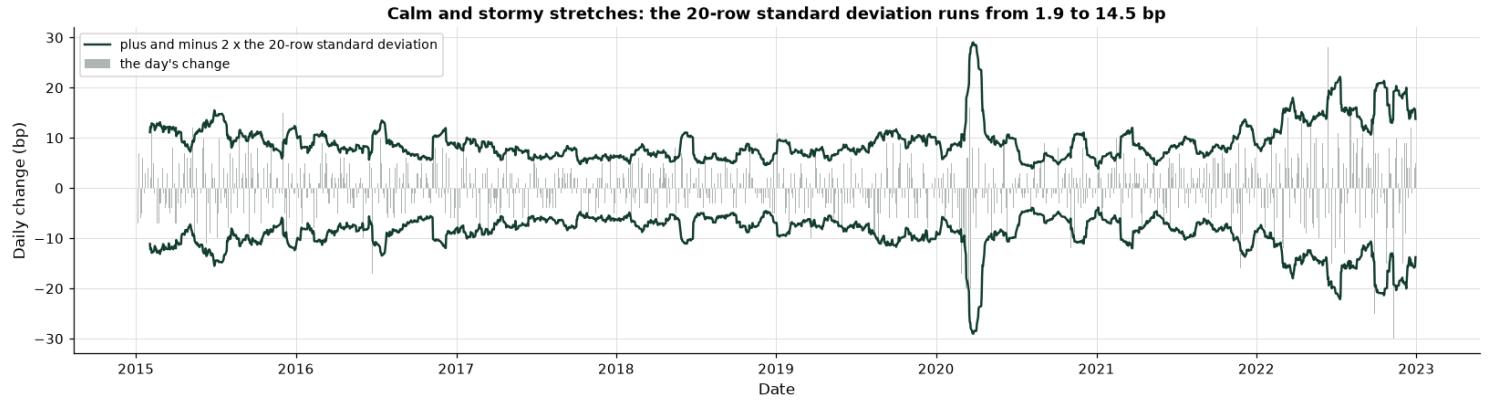
an outlook or a view on rates

Every score is on dates already in the file, from a model fitted on earlier dates.

How volatility clustering works

The pattern	Large moves follow large moves of either sign; quiet days follow quiet days
The catch	A plain ACF of the change cannot see it: positive and negative moves cancel
The fix	Square the change or take its absolute value: the sign is gone
The check	Autocorrelations of the transform, each read against the band
What comes out	The memory of the size, which the change itself does not have

Calm and stormy stretches



Series	Change	Squared	Absolute
10 Yr	0.0225	1.02e-114	2.58e-113
2s10s	0.146	1.68e-60	2.83e-83
IG	0.00237	4.1e-111	1.88e-171

Bars outside the band, of 40: change 4, square 34, absolute 40. The 20-row standard deviation runs from 1.93 bp (5 Jan 2021) to 14.51 bp (26 Mar 2020). Ljung-Box p-values at 10 lags.

The memory is in the size.

How GARCH(1,1) works

The model	The change is mu plus a shock; the size of the shock has its own recursion
The recursion	variance = omega + alpha x (last shock) squared + beta x (last variance)
The fit	Maximum likelihood: the four numbers that make the changes most probable
Persistence	alpha + beta: the share of a variance gap still there one row later
Half-life	ln(0.5) / ln(persistence) rows until half a shock has faded

The coefficients

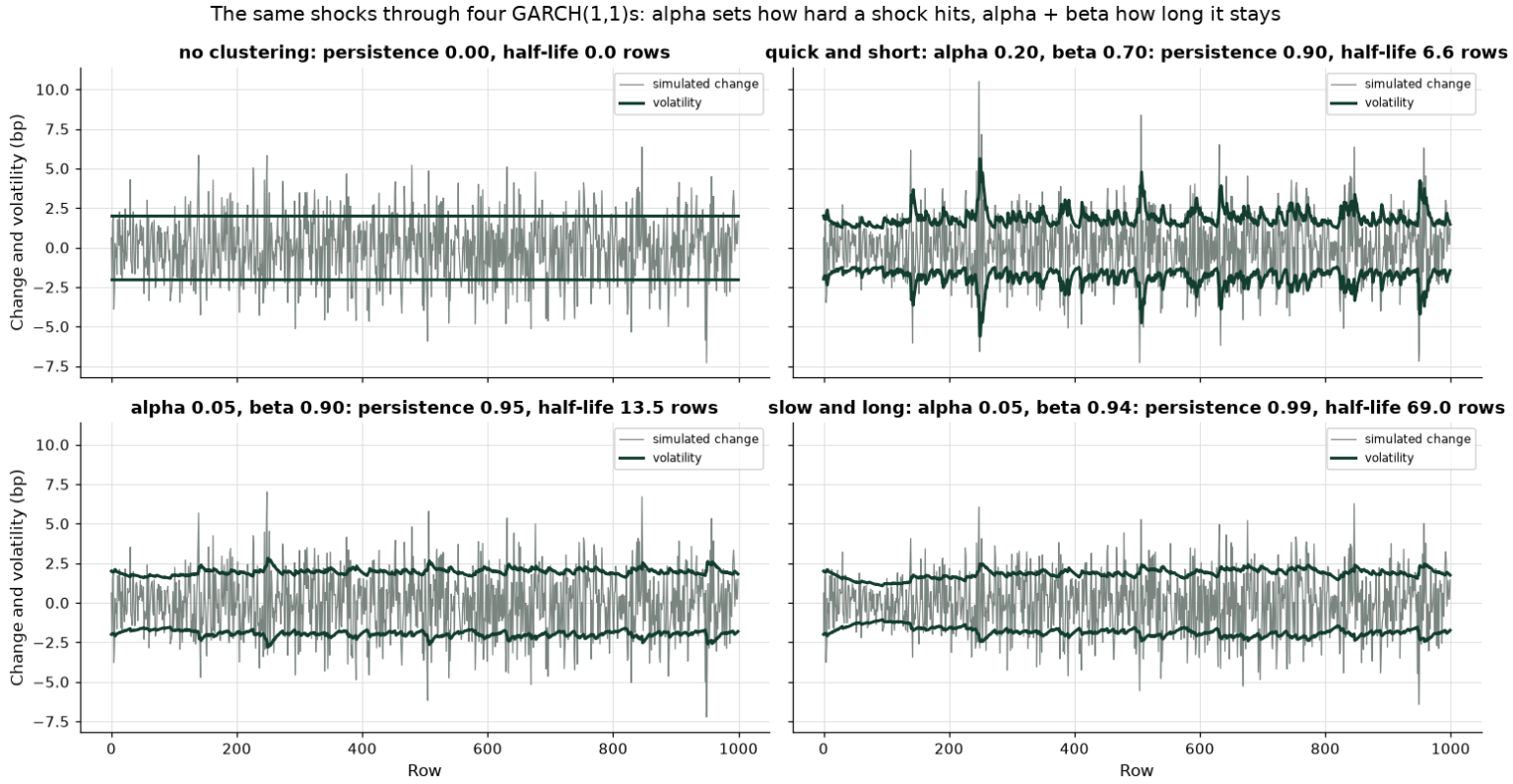
Quantity	Value	Std error
mu (bp)	0.071	0.093
omega	0.467	0.242
alpha	0.066	0.017
beta	0.915	0.025
Persistence	0.981	-
Half-life (rows)	35.950	-
Long-run vol (bp)	4.940	-
Sample sd (bp)	5.030	-

How to read it

A squared shock moves the next variance by alpha of its size; beta of the old variance carries on. The recursion written by hand equals arch's variance on all 2,000 rows to 1e-8: GARCH is one line repeated.

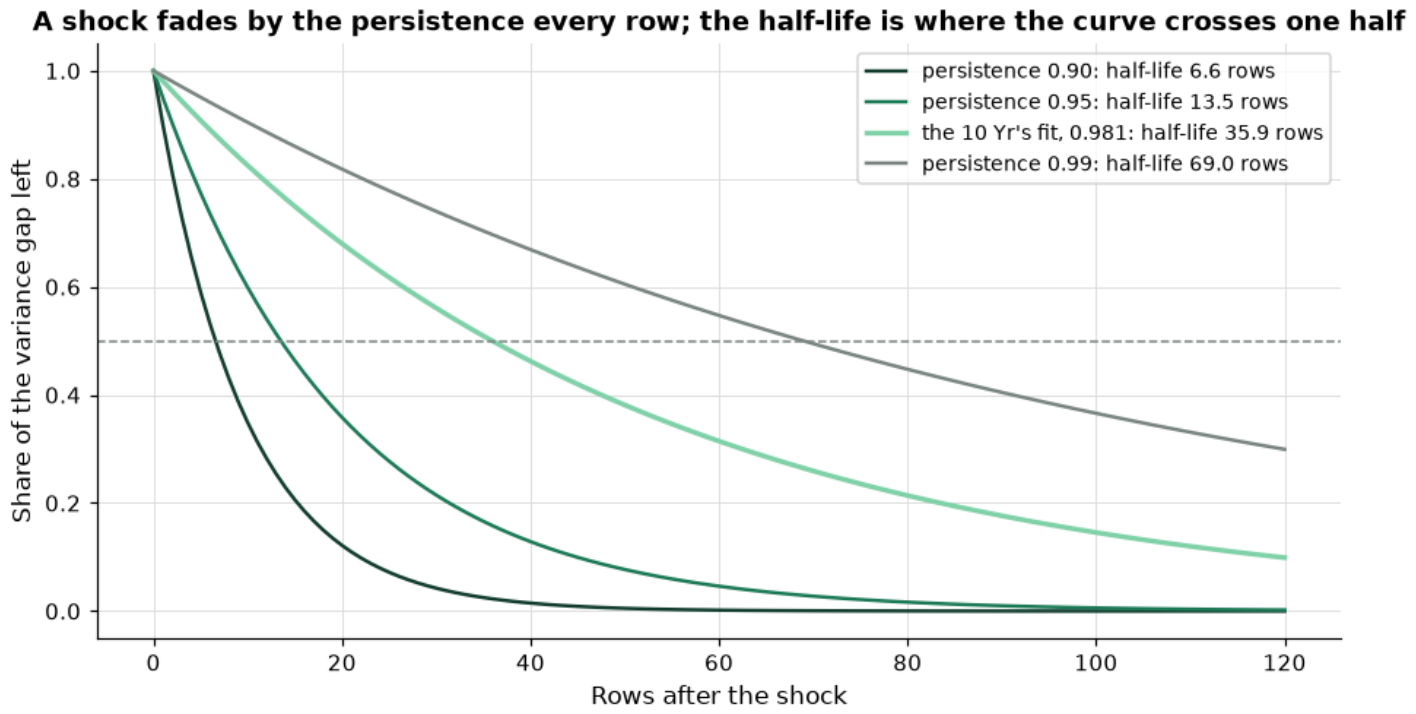
The table shows 3 decimals: the optimiser's last digits move with the machine.

Four GARCH models, the same shocks



Alpha sets how hard a shock hits; alpha + beta sets how long it stays.

How fast a shock fades

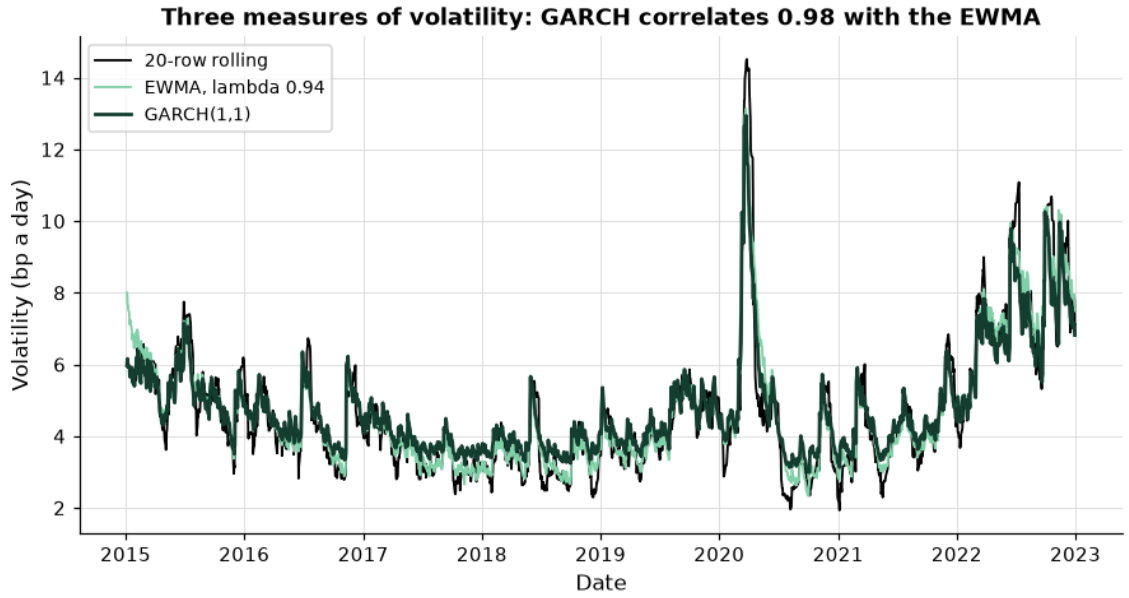


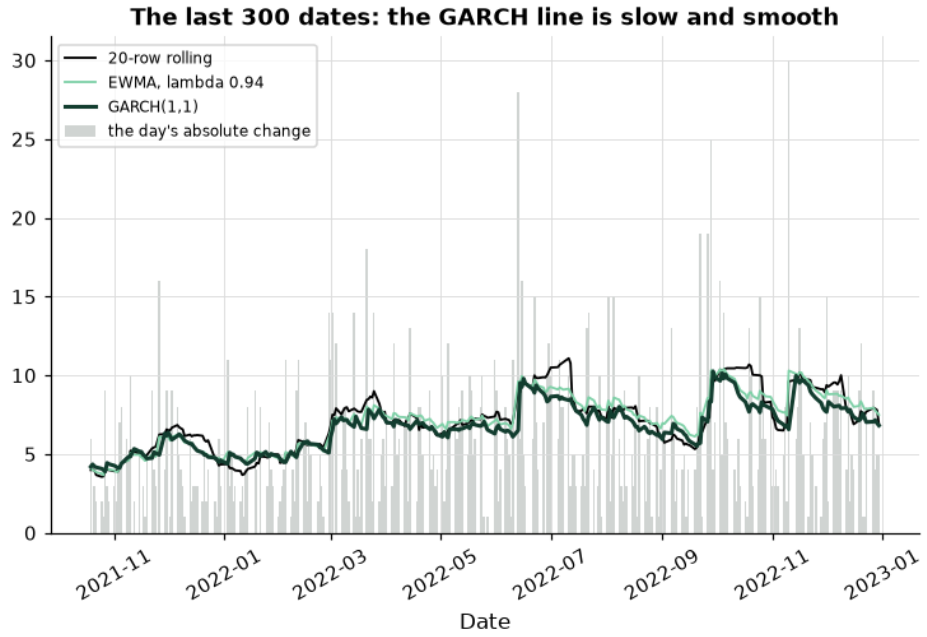
How to read it

Each curve is persistence to the power k. A small change in persistence makes a large change in memory: read persistence first.

The half-life is where the curve crosses one half.

Three volatilities on one chart





All three rise and fall together. The EWMA is a GARCH with omega 0 and persistence 1. The GARCH pulls toward its long-run level between storms: the EWMA has no level to pull toward. A description of the development years, with coefficients fitted on all of them.

Section 7.5 refits fold by fold.

Every setting of arch_model

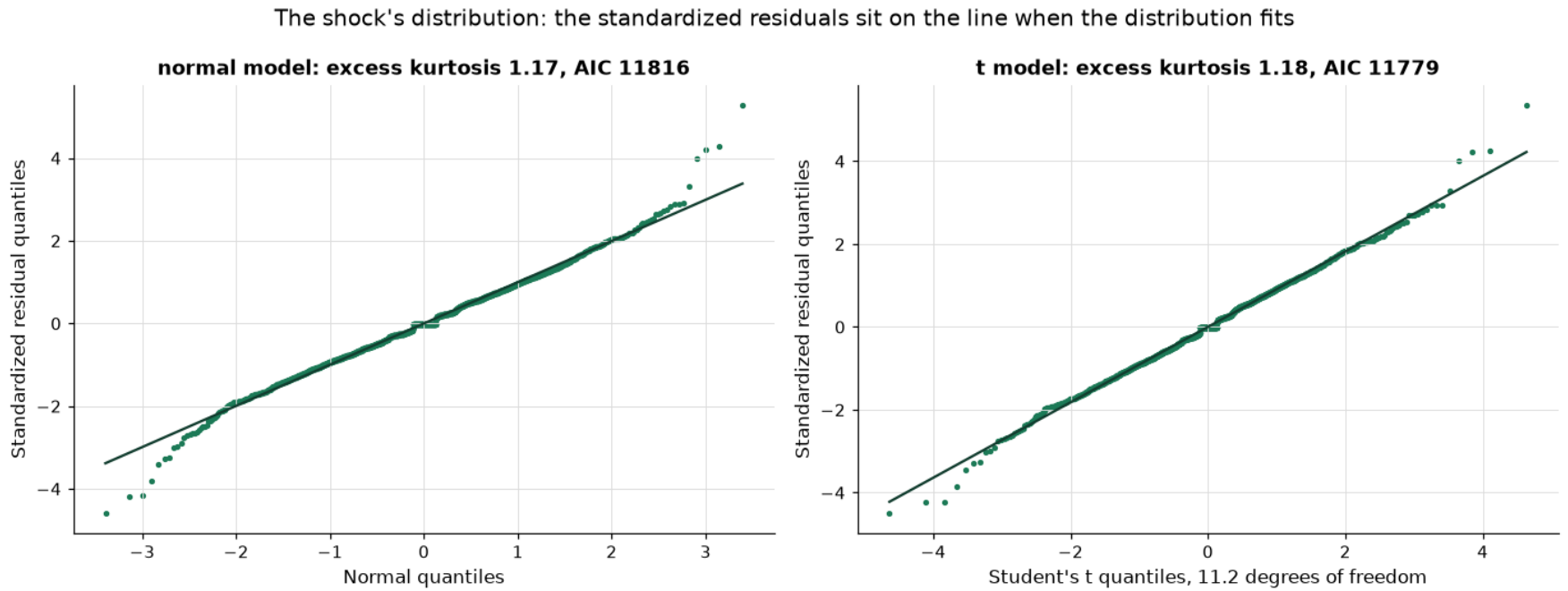
Setting	Default	Tune first
mean	'Constant'	no
vol	'GARCH'	yes
p	1	yes
o	0	yes
q	1	yes
dist	'normal'	yes

dist	AIC	BIC	nu
normal	11816.151	11838.555	-
t	11779.000	11807.004	11.160
skewt	11780.425	11814.030	-
ged	11795.901	11823.906	1.640

The t adds one number, its degrees of freedom: its AIC is lower and its BIC is the lowest. Alpha and beta barely move: the clustering is the same, only the tails change.

The defaults are read from the installed arch with inspect.signature, never typed.

The shock's distribution



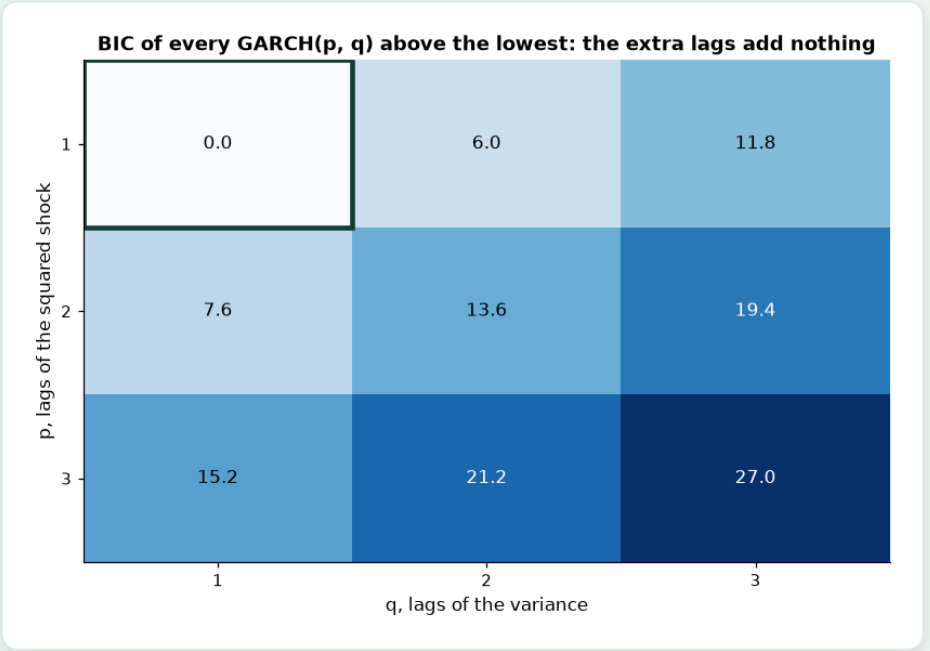
The normal model's tails bend away from the line; the t model's points sit on it. Which tails matter depends on what the volatility is for: a score of size is not much affected.

Standardized residuals sit on the line when the distribution fits.

The variance model and the order

Model	Parameters	AIC	BIC
ARCH(1)	3	12018.85	12035.66
GARCH(1,1)	4	11816.15	11838.56
GJR-GARCH(1,1,1)	5	11818.11	11846.11
EGARCH(1,1)	4	11818.90	11841.30
GARCH(2,1)	5	11818.15	11846.16

Orders that did not converge: 0 of 9. The lowest BIC is at (p, q) = (1, 1).



The course keeps the GARCH(1,1).

How walk-forward GARCH works

Each fold	Fit a normal GARCH(1,1) once, on the training changes only
The test rows	Run the variance recursion on from the last training variance, coefficients fixed
The value	Volatility x 0.7979 for the size of the change: it uses only the rows before the date
The score	MAE against the absolute change, by fold; refit at the next fold
The proof	From day 2: no value may move when the rows from the middle of the fold on are removed

Four rules for the size of the change

10 Yr	Gap (bp)	Ratio	Below
20-row average	-0.0899	0.9578	3
EWMA 0.94	-0.0864	0.9594	3
GARCH(1,1)	-0.0631	0.9704	3

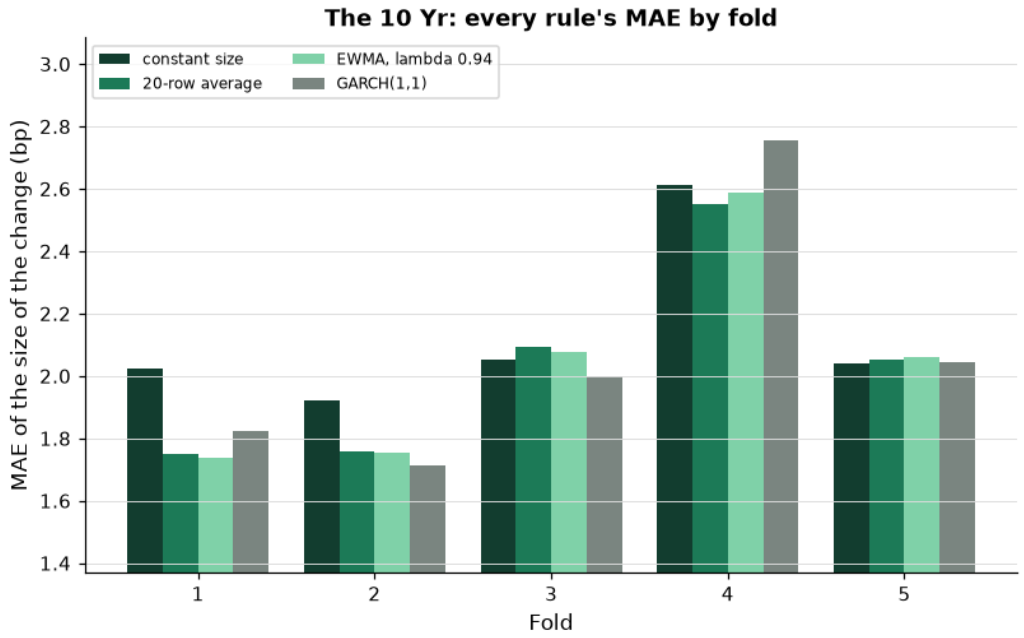
2s10s	Gap (bp)	Ratio	Below
20-row average	-0.0387	0.9741	3
EWMA 0.94	-0.0438	0.9708	3
GARCH(1,1)	-0.0383	0.9745	3

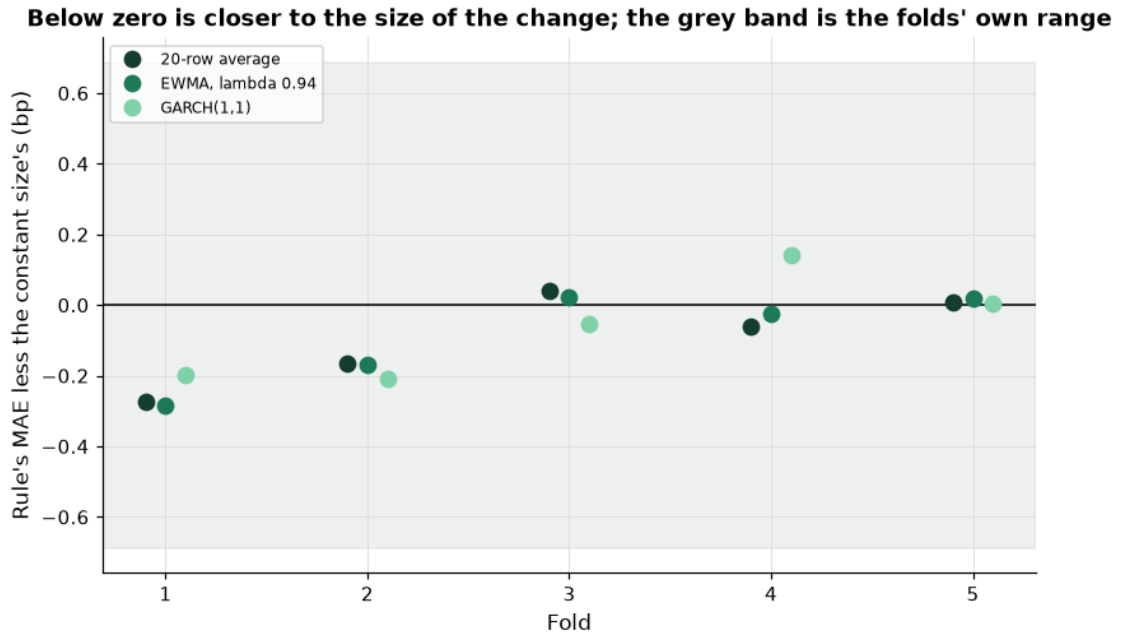
IG	Gap (bp)	Ratio	Below
20-row average	-0.0005	0.9989	3
EWMA 0.94	0.0038	1.0076	2
GARCH(1,1)	-0.0139	0.9725	4

Gap: a rule's MAE for the size of the change less a constant size's, five-fold average. Ratio: the same as a ratio. Below: folds, of 5, below the constant. The constant's own folds differ by 0.6882 bp (10 Yr), 0.4898 (2s10s), and 0.2641 (IG).

Eight of nine are closer than a constant. The GARCH is the closest rule on one series of three.

Fold by fold





The gains are a few percent and the folds do not agree on them. Part of the reason is the target: one day's absolute change is a noisy sample of that day's volatility.

A constant size is the baseline; below zero is closer to the size of the change.

Why the GARCH is not closer

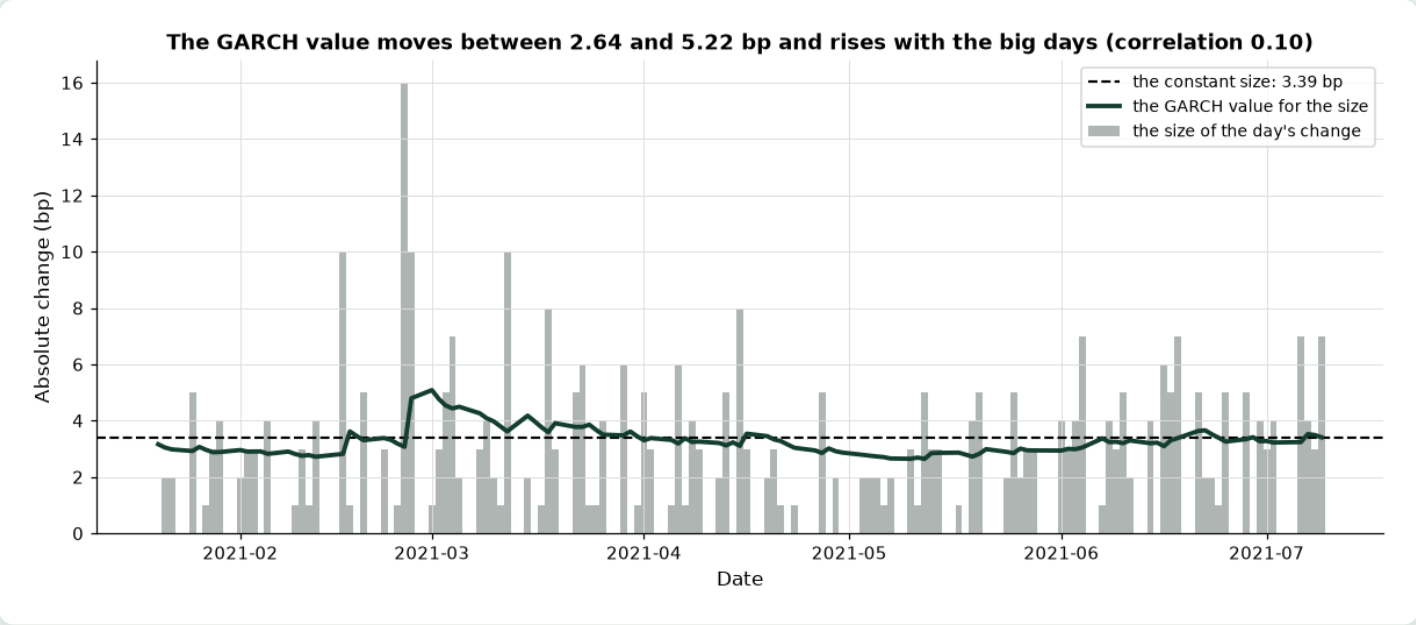
Fold	Fitted on	alpha	beta	Persist.	Long-run (bp)
1	503	0.000	0.996	0.996	3.900
2	755	0.000	0.998	0.998	0.010
3	1,007	0.000	0.997	0.997	3.200
4	1,259	0.009	0.986	0.995	3.910
5	1,511	0.087	0.849	0.936	4.380

What the table says

Folds 1 to 3 found no clustering to fit (alpha near 0): a nearly constant variance. A GARCH also pulls back toward the long-run level it learned in training, which works against it when the test dates are stormier than the training dates.

A fit on a few hundred rows is fragile.

A second yardstick: the log score



Rule	Log score	Gain	Above
constant size	-2.8946	0.0000	0
20-row average	-2.8494	0.0452	2
EWMA 0.94	-2.8273	0.0673	3
GARCH(1,1)	-2.8781	0.0165	3

A log score punishes a volatility that is too small on a big day harder than MAE, so it sees clustering more clearly, and it ranks the EWMA first. The GARCH value knows a little about how big the days tend to be, and nothing about a given day.

Mean Gaussian log score of the signed change, 10 Yr, higher is better.

The level against the size

What is described	Model	Baseline	Ratio
the level of the change	AR(1)	the last value (a change of zero)	1.0028
the size of the change	GARCH(1,1)	a constant size	0.9704

What the models show

The past of a change told nothing about its sign or its amount, and told a little about its size: a few percent in the scores, though overwhelming in the data.

Where they fail

On short windows (alpha near 0 in folds 1 to 3), on tails (a normal is too thin), on persistence (the 2 Yr's is above 1), and on the margin (folds do not agree).

A description of how a rule would have matched changes on later dates inside the file: not a view on rates.

Excel side by side

Excel side by side (tested in Excel)

`=$F$2+$F$3*(B2-$F$1)^2+$F$4*C2`	volatility 6.8197 bp
`=F3+F4`	0.981
`=LN(0.5)/LN(F3+F4)`	36.1 rows
`=SQRT(F2/(1-F3-F4))`	4.9577 bp
`=SUMPRODUCT(ABS(ABS(B3:B2001)-SQRT(2/PI())*D3:D2001))/ROWS(B3:B2001)`	size MAE 2.2991 bp

Coefficients pasted rounded to 3 decimals; within 0.9 bp of arch on every row, 0.0 bp after the first 200. Excel has no GARCH fitter.

Save the day: a rename

1. Commit

The code, then the four coefficients

2. Rename

git mv, then commit

3. Log

A plain log stops at the rename

4. Follow

git log --follow finds the old name's commit

```
Day 7: rename the coefficients file
Day 7: GARCH coefficients
Day 7: tskit with garch_half_life, fit_garch, garch_variance, and walk_forward_garch
```

A rename is a change Git can follow, but only when you ask.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the GARCH is fitted to the levels, not the changes: its average volatility is 21.2 bp for a series whose daily changes have a standard deviation of 3.0 bp.

Code that runs is not the same as code that is right.

What you can do now

- The size of a change has memory its level does not
- GARCH(1,1): one recursion, persistence, half-life
- Every setting of arch_model: distribution, variance model, order
- Walk-forward: a few percent closer than a constant size
- The GARCH was the closest rule on one series of three

Week 2, Day 3 Lag features and gradient boosting